Karatsuba Algorithm

$x = 1234$ (a = 12, b = 34)

$= 1200 + 34$

$= 12 \times 10^2 + 34$

$\rightarrow a \times 10^{\frac{n}{2}} + b$

$y = 5678$ (c = 56, d = 78)

$= 5600 + 78$

$= 56 \times 10^2 + 78$

$\rightarrow c \times 10^{\frac{n}{2}} + d$

$n \rightarrow$ no. of digits $= 4$

$$xy = \left(a \times 10^{\frac{n}{2}} + b\right)\left(c \times 10^{\frac{n}{2}} + d\right)$$

$$= ac \times 10^n + ad \times 10^{\frac{n}{2}} + bc \times 10^{\frac{n}{2}} + bd$$

$$= \underset{1}{ac} \times 10^n + \left(\underset{2}{ad} + \underset{3}{bc}\right) \times 10^{\frac{n}{2}} + \underset{4}{bd} \quad \text{------} (1)$$

$4 \rightarrow O(n^2)$

$$(a+b)(c+d) - ac - bd$$
$$= ac + ad + bc + bd - ac - bd$$
$$= ad + bc$$

From (1) $\rightarrow$

$$\underset{1}{ac} \times 10^n + \left\{\underset{2}{(a+b)(c+d)} - ac - \underset{3}{bd}\right\} \times 10^{\frac{n}{2}} + bd$$

$\downarrow$

$O(n^{1.59})$